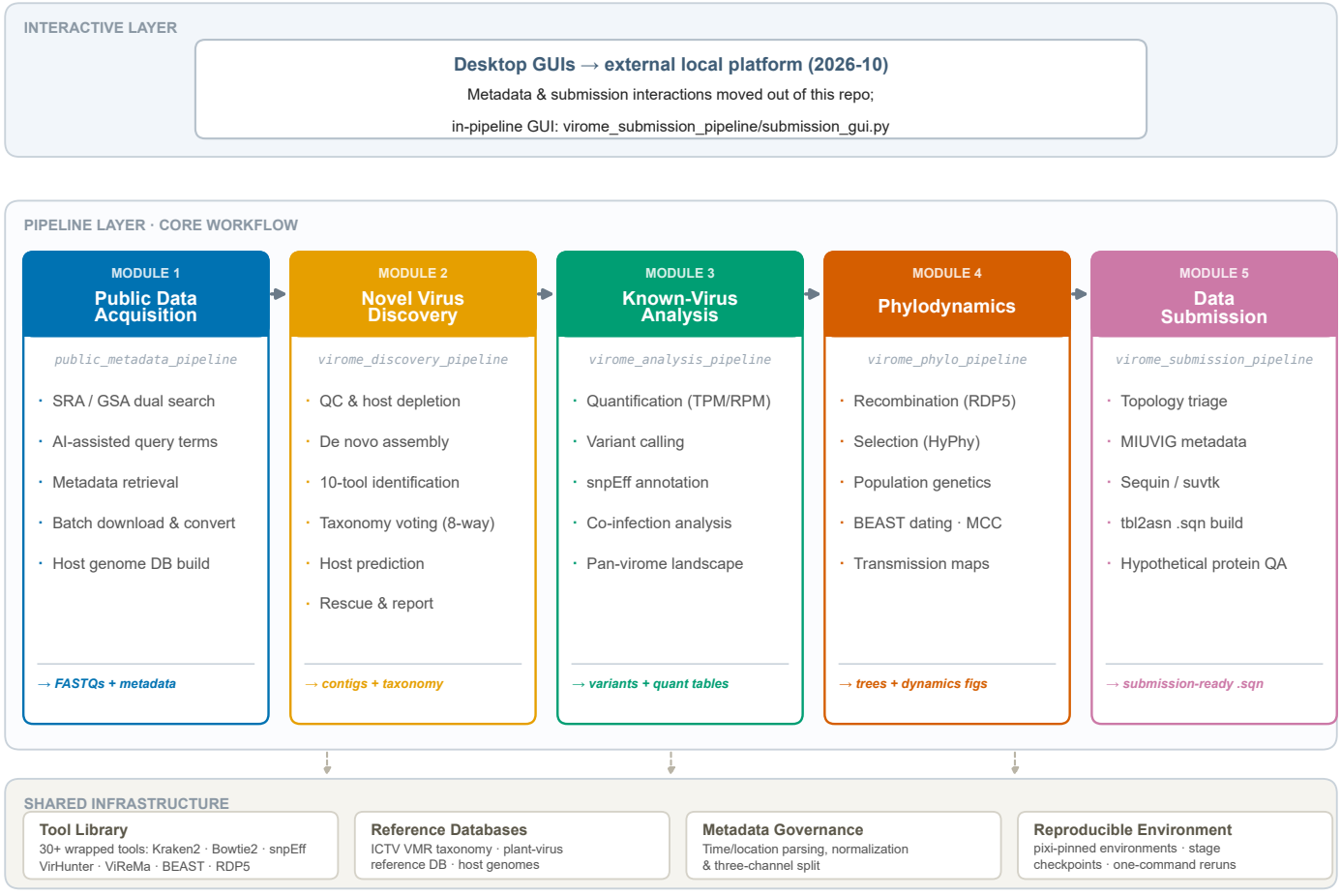

INTERACTIVE LAYER
Desktop GUIs → external local platform (2026-10)
Metadata & submission interactions moved out of this repo;
in-pipeline GUI: virome_submission_pipeline/submission_gui.py
PIPELINE LAYER · CORE WORKFLOW
MODULE 1
Public Data Acquisition
public_metadata_pipeline
SRA / GSA dual search
AI-assisted query terms
Metadata retrieval
Batch download & convert
Host genome DB build
→ FASTQs + metadata
MODULE 2
Novel Virus Discovery
virome_discovery_pipeline
QC & host depletion
De novo assembly
10-tool identification
Taxonomy voting (8-way)
Host prediction
Rescue & report
→ contigs + taxonomy
MODULE 3
Known-Virus Analysis
virome_analysis_pipeline
Quantification (TPM/RPM)
Variant calling
snpEff annotation
Co-infection analysis
Pan-virome landscape
→ variants + quant tables
MODULE 4
Phylodynamics
virome_phylo_pipeline
Recombination (RDP5)
Selection (HyPhy)
Population genetics
BEAST dating · MCC
Transmission maps
→ trees + dynamics figs
MODULE 5
Data Submission
virome_submission_pipeline
Topology triage
MIUVIG metadata
Sequin / suvtk
tbl2asn .sqn build
Hypothetical protein QA
→ submission-ready .sqn
SHARED INFRASTRUCTURE
Tool Library
30+ wrapped tools: Kraken2 · Bowtie2 · snpEff
VirHunter · ViReMa · BEAST · RDP5
Reference Databases
ICTV VMR taxonomy · plant-virus
reference DB · host genomes
Metadata Governance
Time/location parsing, normalization
& three-channel split
Reproducible Environment
pixi-pinned environments · stage
checkpoints · one-command reruns